



TARGET CAT 2026



CAT 2026 – QUANTS

Algebra

Equations L1

Lec - 01

By– VINIT KAKRIYA SIR





Algebra :



Sr. No.	Topics
1	Linear Equations ✓
2	Quadratic Equations ✓
3	Exponents ✓
4	Inequalities ✓
5	Logarithm ✓
6	Function Basics ✓
7	Modulus Function ✓

TOPICS *to be covered*

- 1) Linear Equations with 'One & Two Variables'
- 2) 'Types of solution' in Linear Equation
- 3) Linear Equations with 'Three Variables'
- 4) Problems for Practice & CAT PYQs



Linear Equation :



$$3x + 5 = 8$$

$$3x = 3$$

$$\boxed{x = 1}$$

$$2x + 5y = 10$$

$$2x + 3y + 5z = 20$$



Solution of Linear Equation :

- With Single Variable :

$$3x - \frac{8}{3} = \frac{7}{3}$$

$$3x = \frac{7}{3} + \frac{8}{3}$$

$$3x = \frac{15}{3}$$

$$3x = 5$$

$$x = \frac{5}{3}$$

Solution of Linear Equation :

- With Two Variables :
- Method of Substitution :

1 to 30
Tables

$$\begin{array}{lcl}
 2x + 5y = 18 & | & 3x - y = 10 \\
 & & \boxed{3x - 10 = y} \\
 2x + 5(3x - 10) = 18 & & \\
 2x + 15x - 50 = 18 & & \\
 17x = 68 & & \\
 \boxed{x = 4} & & \\
 & & 3(4) - 10 = y \\
 & & 12 - 10 = y \\
 & & \boxed{2 = y}
 \end{array}$$

Solution of Linear Equation :

- With Two Variables :
- ## 2. Method of Elimination :

$$\begin{array}{rcl}
 2x + \cancel{5y} = 18 & | & \boxed{3x - y = 10} \times 5 \\
 & & 15x - \cancel{5y} = 50
 \end{array}$$

(+)

$$17x = 68$$

$$\boxed{x = 4}$$

$$3x - y = 10$$

$$3(4) - y = 10$$

$$12 - 10 = y$$

$$\boxed{2 = y}$$

① coe same

② Sign

same $\rightarrow$ subtraction
opposite $\rightarrow$ Addition

QUESTION-1



#Q. The father's age is six times his son's age. Four years ^(after) hence, the age of the father will be four times his son's age. The present age (in years) of the son is,

let, Present $\xrightarrow{+4 \text{ yr}}$ 4 yrs After

Son = x

Father = $6x$

Son = $(x + 4)$

Father = $(6x + 4)$

$\text{Father} = 4(\text{Son})$

$$6x + 4 = 4(x + 4)$$

$$6x + 4 = 4x + 16$$

$$2x = 12$$

$x = 6$

A 6 ✓

B 4

C 5

D 7

QUESTION- 2



#Q. Ten years ago, mother was 12 times as old as her daughter and ten years, hence she will be twice as old as her daughter will be. Find the present age of mother.

A 36

B 34

C 30

D 42

Handwritten solution:

Timeline: 10 yrs Ago $\xrightarrow{+10}$ Present $\xrightarrow{+10}$ 10 yrs After

Let, $D = x$
 $M = 12x$

Present:
 $D = (x + 10)$
 $M = (12x + 10)$

10 yrs After:
 $D = (\underline{x + 10} + \underline{10}) = (x + 20)$
 $M = (\underline{12x + 10} + \underline{10}) = (12x + 20)$

Mother = 2 (Daughter)

$$12x + 20 = 2(x + 20)$$
$$12x + 20 = 2x + 40$$
$$10x = 20$$
$$x = 2$$

Substituting $x = 2$ into the present age of mother:
 $M = 12(2) + 10$
 $= 34$

QUESTION- 3

$$B + C = 20$$

$$x + C = 20$$

$$C = (20 - x)$$

let, $B \rightarrow x$ shares



#Q. Stocks A, B and C are priced at rupees 120, 90 and 150 per share, respectively. A trader holds a portfolio consisting of 10 shares of stock A, and 20 shares of stocks B and C put together. If the total value of her portfolio is rupees 3300, then the number of shares of stock B that she holds, is 15 [CAT 2025: Slot 1]

	<u>Price/share</u>	<u>Qty of shares</u>
A	120	10
B	90	<u>x</u>
C	150	<u>(20-x)</u>

$$\text{Value} = (\text{Price/share}) \times (\text{Qty of shares})$$

$$T \text{ value} = 3300$$

$$V_A + V_B + V_C = 3300$$

$$120(10) + 90(x) + 150(20-x) = 3300$$

$$1200 + 90x + 3000 - 150x = 3300$$

$$4200 - 60x = 3300$$

$$900 = 60x$$

$$\frac{90}{6} = x$$

$$15 = x$$

QUESTION- 4



#Q. 3 tables and 2 chairs cost Rs. 1900 and 2 tables and 4 chairs cost Rs. 1800. Find the cost of table and a chair. The total price of 10 chairs and 1 table is:

Let, Price / Table = Rs. x $\rightarrow$ 500
Price / chair = Rs y $\rightarrow$ 200

A 3200

B 2700

C 3000

D 2500

$$\begin{aligned} 3x + 2y &= 1900 \\ 2x + 4y &= 1800 \\ \hline x + 2y &= 900 \\ 2y &= 900 - x \\ 3x + 900 - x &= 1900 \\ 2x &= 1000 \\ x &= 500 \\ 2y &= 900 - 500 \\ 2y &= 400 \\ y &= 200 \end{aligned}$$

$$\begin{aligned} 10c \& 1T \\ \div 2 &= 10y + x \\ &= 10(200) + 500 \\ &= \underline{\underline{2500}} \end{aligned}$$

QUESTION- 5



#Q. A man's age is three times the sum of the ages of his two sons. After 5 years, his age will be twice the sum of his two son's age. Find the age of the man.

Age of Man = m

Age of first son = x

Second son = y

Present

$$m = 3(x + y)$$

$$m = 3x + 3y$$

$$\begin{aligned} m &= 3(x + y) \\ &= 3(15) = 45 \end{aligned}$$

After 5 yrs

$$\text{Man} = (m + 5)$$

$$\text{first son} = (x + 5)$$

$$\text{Second son} = (y + 5)$$

$$\text{Man} = 2[(\text{Son})_1 + (\text{Son})_2]$$

$$(m) + 5 = 2[x + 5 + y + 5]$$

$$3x + 3y + 5 = 2(x + y + 10)$$

$$3x + 3y + 5 = 2x + 2y + 20$$

$$x + y = 15$$

QUESTION- 6



#Q. If 2 is added to the numerator and denominator it becomes $9/10$ and if 3 is subtracted from the numerator and denominator it become $4/5$. Find the numerator of the original fraction.

original fraction = $\frac{x}{y}$

(I)

$$\frac{(x+2)}{(y+2)} = \frac{9}{10}$$

$$10x + 20 = 9y + 18$$

$$10x = 9y - 2$$

$$8y + 6 = 9y - 2$$

$$8 = y$$

(II)

$$\frac{(x-3)}{(y-3)} = \frac{4}{5}$$

$$5x - 15 = 4y - 12$$

$$5x = 4y + 3$$

$$10x = 8y + 6$$

$$10x = 8(8) + 6$$

$$10x = 70$$

$$x = 7$$

A 8

B 7

C 1

D 15



Types of Solution in Linear Equation :

$$a_1x + b_1y = c_1$$

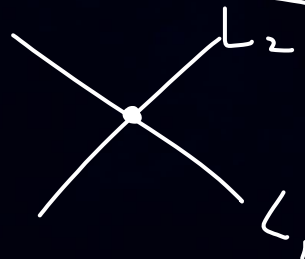
$$a_2x + b_2y = c_2$$

_____ L_1

_____ L_2

① Unique Soln

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$



② Infinit Soln

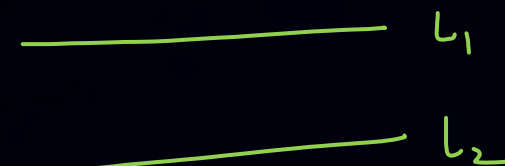
$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$



(overlapping Soln)

③ No Soln

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$



(parallel lines)



Types of Solution in Linear Equation :

If there are two linear equations,

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

Unique Solution

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

(overlapping Lines)
Infinite Solution

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

(Parallel Lines)
No Solution

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

QUESTION- 7



#Q. For what value of k , the lines with equation $(k - 1)x + y - 2 = 0$ and $(k - 2)x - 3y + 1 = 0$ are parallel to each other

$$\underline{(k-1)}x + y = 2$$

$$\underline{(k-2)}x - 3y = -1$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2}$$

$$\frac{(k-1)}{(k-2)} = \frac{1}{-3}$$

$$-3(k-1) = k-2$$

$$-3k + 3 = k - 2$$

$$5 = 4k$$

$$\boxed{\frac{5}{4} = k}$$

A $k = \frac{4}{5}$

B $k = \frac{5}{4}$

C $k = \frac{-4}{5}$

D None of these

QUESTION- 8



#Q. Let k be a constant. The equations $kx + y = 3$ and $4x + ky = 4$ have a unique solution if and only if
[CAT 2020 : Slot 3]

$$\begin{aligned}(\pm 2)^2 &= 4 \\ (-2)^2 &= 4\end{aligned}$$

$$\begin{aligned}kx + 1y &= 3 \\ 4x + ky &= 4\end{aligned}$$

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

Unique Solⁿ

$$\frac{k}{4} \neq \frac{1}{k}$$

$$k^2 \neq 4$$

$$k^2 \neq (\pm 2)^2$$

$$k \neq \pm 2$$

A $k = \pm 2$

B $k \neq 2$

C $k \neq -2$

D $k \neq \pm 2$

QUESTION- 9



#Q. Find the value of ' $\underline{\underline{p + q}}$ ' if the pair of equations

$$\begin{array}{r} -2 + 6 \\ \boxed{4} \end{array}$$

$$2x + py = 8$$

$$qx - 6y = 24$$

has infinite solutions.

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\frac{2}{9} = \frac{p}{-6} = \frac{8}{24}$$

$$\frac{2}{9} = \frac{p}{-6} = \frac{1}{3}$$

$$\frac{2}{9} = \frac{1}{3}$$
$$\boxed{9 = 6}$$

$$\frac{p}{-6} = \frac{1}{3}$$
$$3p = -6$$
$$\boxed{p = -2}$$



$$y = 3$$

QUESTION-10

$$\frac{462}{11} = 42$$

$$42 + 13 = \underline{546}$$



#Q. Cost of 8 Pencils, 5 Pens and 3 erasers is Rs. 111. Cost of 9 Pencils, 6 Pens and 5 erasers is Rs. 130. Cost of 16 Pencils, 11 Pens and 3 erasers is Rs. 221. What is the cost of 39 Pencils, 26 Pens and 13 erasers?

$$\begin{aligned} \text{let, cost/Pencil} &= a \\ \text{cost/Pen} &= b \\ \text{cost/eraser} &= c \end{aligned}$$

$$\begin{aligned} + 8a + 5b + 3c &= 111 \\ + 9a + 6b + 5c &= 130 \\ + 16a + 11b + 3c &= 221 \end{aligned}$$

$$\underline{33a + 22b + 11c = 462}$$

$$\underline{3a + 2b + 1c = 42}$$

$$39a + 26b + 13c = 546$$

$$\boxed{39a + 26b + 13c}$$

?

- 11

x 13

A 316

B 546

C 624

D 482

QUESTION- 11



#Q. Cost of 4 pens, 6 Note-books and 9 files is Rs. 305. Cost of 3 pens, 4 note-books and 2 files is Rs. 145. What is the cost of 5 pens, 8 note-books and 16 files?

$$\begin{array}{l}
 \text{Cost/pen} \rightarrow a \\
 \text{Cost/NB} \rightarrow b \\
 \text{Cost/file} \rightarrow c
 \end{array}
 \quad \times 2 \quad
 \boxed{4a + 6b + 9c = 305} \quad | \quad 3a + 4b + 2c = 145$$

$$8a + 12b + 18c = 610$$

(-)

A 465

$$\underline{5a} + \underline{8b} + \underline{16c} = ?$$

B 456

$$5a + 8b + 16c = 610 - 145$$

$$\boxed{5a + 8b + 16c = 465}$$

C 440

D None of these

QUESTION- 12



#Q. If $a - 6b + 6c = 4$ and $6a + 3b - 3c = 50$, where a , b and c are real numbers, the value of $2a + 3b - 3c$ is [CAT 2025: Slot 1]

$$\begin{array}{l} a - \cancel{6b} + \cancel{6c} = 4 \\ \boxed{6a + \cancel{3b} - \cancel{3c} = 50} \times 2 \\ \hline 12a + \cancel{6b} - \cancel{6c} = 100 \\ \xrightarrow{(+)} \\ 13a = 104 \\ a = \frac{104}{13} = \boxed{8} \end{array} \quad \begin{array}{l} \boxed{2a + \cancel{3b} - \cancel{3c}} \\ = 2(8) + 2 \\ = 16 + 2 = \underline{\underline{18}} \end{array}$$
$$\begin{array}{l} 6a + 3b - 3c = 50 \\ 6(8) + 3b - 3c = 50 \\ \xrightarrow{(-)} \\ \boxed{3b - 3c = 2} \end{array}$$

A 18 ✓

B 20

C 15

D 14



2 Mins Summary

- 1) Linear Equations with 'One & Two Variables'
- 2) 'Types of solution' in Linear Equation
- 3) Linear Equations with 'Three Variables'
- 4) Problems for Practice & CAT PYQs



*Thank
You*